

Kryptos K4

An approach

1. Definition of terms

Complementary letters

(regarding to the Vigenère encryption)

Two letters whose shift values sum up to 26, e.g.:

- T (19) and H (7) in the standard alphabet
- U (20) and G (6) in the standard alphabet
- Q (20) and S (6) in the Kryptos alphabet
- U (21) and O (5) in the Kryptos alphabet

The Model

This term refers to the Kryptos model, which was presented by Ed Scheidt in 2015.

2. Double letters

Double letters seem to play an important role in Kryptos:

- The misspellings in K1 and K2 mark the first letter of a double letter (LL, UU).
- There are noticeably many double letters in K4 and some of them have the same distance to each other.
- The missing ciphertext letter in the model (N) marks a double letter too (IAK**NN**VEK).

First, let's take a closer look at the structure of double letters in K4:

OBKRUOXOGHULBSOLIF BB WFLRV QQ PRNGK SS OTWTQSJQ SS EKZZ WATJKLUDIAWINFBNYPVTTMZFPKWGDKZXTJCDIGKUHUAUEKCA R

(Double letters in K4)

As already known, there are three raised letters on the sculpture (ENDyaHr).

If you follow the easiest interpretation of this hint by changing the position of 3 ciphertext letters in K4, you get the following ciphertext:

OBKRUOXOGHULBSOLIFBBWFLRVQQPRNGKSSOTWTQSJQSSEKZZ
WATJKLUDIAWINFBNYPVTTMZFPKWGDKZXTJCDIGKUHUAAUEKCA
R

OBKRUOOGHULBSOLIFBBWFLRVQQPRNGKSSOTWWTQSJQSSEKZZ
WATJKLUDIAAAWINFBNYPVTTMZFPKGDKZXXTJCDIGKUHUUUEKCA
R

(Changing the position of three K4 ciphertext letters)

The double letters QQ, WW, ZZ, AA, TT, XX and UU now have exactly the same distance to each other. The appearance of these double letters could therefore be a property of the encryption system.

There is something remarkable about the double letter NN in the model too.

The very short Vigenère key RUG is not a meaningful word like PALIMPSEST or ABSCISSA in Kryptos. So I was wondering if RUG is really the right key or maybe just a „shifted version“ of the right key, which would point to a second layer of encryption with a monoalphabetic key.

Among all 25 shifted versions of the key RUG, there is also the reversal of the plaintext word THE (RUG – NNN = EHT), with the key letter N (ROT13) being the same as the double letter NN in the ciphertext. Furthermore, the plaintext word THE is located at the same position as the first missing N in the ciphertext.

So, assuming that the correct key is actually EHT, the two-layer encryption process of the last plaintext line would look like this:

WHTHOUTTHEKEY	Plaintext
EHT EHT EHT EHT	Layer 1 (polyalphabetic)
AOMLVNXXAAIRXC	Intermediate plaintext
NNNNNNNNNNNNNN	Layer 2 (monoalphabetic)
NBZYIAKNNVEKP	Ciphertext

(A two-layer encryption in the model?)

If this is correct, the plaintext bigram TH was encrypted with its reverse version on the first layer, which always results in a double letter. But there is something more to it. Since T and H are complementary letters, the resulting double letter is the first letter of the alphabet (A). That is why the key letter on the second layer is the same as the double letter in the ciphertext.

Interestingly, there is the string XAAIRX at this position in the intermediate plaintext, but I'm not sure if this all has anything to do with the encryption system.

3. K4 - A potential start of the plaintext

There is some evidence that a letter is perhaps missing at the beginning of K4. In this approach, the ciphertext starts with an L and the X from OXO is missing due to the change in the position of 3 ciphertext letters as described in the last chapter:

```
LOB KRUOOGHU LBS OLIFBBWF
```

(The start of K4)

If you decrypt the first 3 letters LOB with KEY, you get LQS as result, which is the key to decrypt the following 8 letters KRUOOGHU. The same system works for LBS and OLIFBBWF:

```
LOB KRUOOGHU LBS OLIFBBWF  
KEY LQSLQSLQ KEY LWTLWTLW  
LQS CAIHEAWR LWT HQEUETSI
```

(A readable plaintext emerges from the mist)

At this point it should be briefly mentioned that the key LQS is just a monoalphabetically substituted version of the key RUG, with the Kryptos alphabet used as the key:

ABCDEFGHIJKLMNOPQRSTUVWXYZ Plaintext alphabet
KRYPTOSABCDEFGHIJLMNQUVWXZ Ciphertext alphabet

RUG → LQS

(RUG becomes LQS)

However, the two blocks LOB and LBS also produce a readable plaintext when decrypted with the key XYY:

L OB KRUOOGHU LBS OLIFBBWF
X YY LQSLQSLQ XYY LWTLWTLW
N PS CAIHEAWR NST HQUETS I

(A readable plaintext emerges from the mist)

The plaintext bigrams AI, QE and TS seem to be reversed:

N PS CAIHEAWR NST HQUETS I
N PS CIAHEAWR NST HEQUEST I
N PS CIA H EAWRNS THE QUEST I

(A readable plaintext emerges from the mist)

It took me a while to figure out that the letters EAWRNS are probably supposed to be the word ANSWER:

N PS CIA H E AWRNS THE QUESTI

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N PS CIA H E ANSWR THE QUESTI

A +3 = N

N +1 = S

S -3 = W

W +1 = R

(Transposition of the word answer)

It looks like the fifth letter of answ(e)r was taken out and put in front of the word before the transposition was performed.

So it can be assumed that the last three letters are either ONS or ONX.

Unfortunately, the system does not work for the next eleven letters LRV QQPRNGKS and the meaning of the first letter N and the letter H (CIAH) remains unclear.

But there is hope because I also found a key to decrypt QQPRNGKS.

4. K4 - QQPRNGKS = NORT(H)EAST

A key consisting of three letters is also needed to decrypt QQPRNGKS, but this time I had to recover this key by performing a brute force attack because the letters LRV don't seem to contain any key information. So the key is most likely VGY:

```
QQPRNGKS S  
VGYVGYVG Y  
XAROSETN T
```

(Key VGY - Three letters again)

The second step is a transposition that is very similar to the transposition of the word ANSWER:

```
XAROSETN T  
76328541  
NORTEAXS T
```

```
N -4 = O  
O -1 = R  
R +4 = T  
T -1 = E  
E -4 = A  
A -1 = X  
X +4 = S
```

(Transposition of the word northeast)

In contrast to the transposition of the word ANSWER, this transposition works from right to left. So is it possible that this part of the plaintext is just upside down?

The position of the missing letter is also the same as in answer:

ANSW(E)R NORT(H)EAXST (Missing letters)

However, a lot of important questions remain unanswered:

- Why does the system „interrupt“
- What's about the letter X?
- Where is the missing letter H?
- Is there a second layer as the model suggests?
- What's about the double letters?

5. K4 - Complementary letters

The previous chapters indicate that a transpositional cipher was used in some way. But so far we have encountered two different systems:

- Reversed bigrams (TH**Q**EU)
- More complex transposition (ANSWER, NORTHEAST)

It is very unlikely that two different systems were used. So the reversed bigrams are possibly more a hint than a transposition cipher.

Then I found a kind of system which has something to do with complementary letters and reversed bigrams. The first reversed plaintext bigram AI is encrypted with the complementary key letters Q and S:

KRU00GHU
LQSLQSLQ
CAIHEAWR

(AI + QS = RU)

But there is a little secret. You don't have to know the key letters QS to decrypt this ciphertext bigram. Just decrypt this bigram with its reversed version and use the reversed version of the result as a key:

RU - UR = SQ
RU - QS = AI

(Decryption without knowing the key)

Unfortunately, this system only works if one of the two distances between the two plaintext letters in the alphabet is divisible by 3. But I don't think that's a coincidence because there is a hint to that in K1:

YQTQUXQBQVYUVLLTREVJYQTM KYR DMFD

(K1 hints to a reversed bigram after an K)